

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence**

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage:
35%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I ____N.A.Sriharan/1911260036____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

ANSWER:

Problem formulation (CSP)

- **Variables:** D1, D2, D3 (one per doctor)
- **Domain:** {Morning (M), Afternoon (A), Night (N)}, ordered $M < A < N$
- **Constraints:**
 - C1: $D1 \neq \text{Night}$
 - C2: $\text{shift}(D2) < \text{shift}(D3)$
 - C3: all different (only one doctor per shift, and each doctor gets exactly one shift)
 - C4: $D3 \neq \text{Morning}$

Variables are assigned in the order $D1 \rightarrow D2 \rightarrow D3$, and values are tried in the order M, A, N. After every assignment the constraints that involve only assigned variables are checked; on a violation the algorithm backtracks.

Step-by-step search trace Step	Partial assignment	Constraint check	Result
1	D1 = Morning	C1 ✓ (not Night); C3 ✓	Accept
2	D1 = M, D2 = Morning	C3 X (Morning already used by D1)	Reject → next value

3	D1 = M, D2 = Afternoon	C3 ✓	Accept
4	D1 = M, D2 = A, D3 = Morning	C4 X (D3 cannot work Morning); C3 X	Reject → next value
5	D1 = M, D2 = A, D3 = Afternoon	C3 X (Afternoon used by D2); C2 X (A is not before A)	Reject → next value
6	D1 = M, D2 = A, D3 = Night	C1 ✓; C2 ✓ (A < N); C3 ✓; C4 ✓; all doctors assigned	SOLUTION FOUND
Step	Partial assignment	Constraint check	Result
7	D1 = M, D2 = Night	C3 ✓	Accept
8	D2 = N, D3 = Morning / Afternoon / Night	Morning X (C4); Afternoon X (C2: N is not before A); Night X (C3)	Backtrack to D1
9	D1 = Afternoon, D2 = Morning	C1 ✓; C3 ✓	Accept
10	D1 = A, D2 = M, D3 = Night	Morning X (C4), Afternoon X (C3), Night ✓ (C2: M < N)	2nd solution
11	D1 = A, D2 = Night, D3 = any	Morning X (C4), Afternoon X (C3), Night X (C3)	Backtrack
12	D1 = Night	C1 X (D1 cannot work Night)	Pruned

Note: (D1 = Afternoon, D2 = Morning, D3 = Night) is also valid, so the problem has exactly two solutions. Backtracking returns the first one it meets.

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

ANSWER:

Algorithm

1. Start from the **Start (S)** node.
2. Calculate the Manhattan distance $\|h(n)\|$ for each possible node.
3. Calculate $\|f(n)=g(n)+h(n)\|$.
4. Add the valid nodes to the priority queue according to their $\|f\|$ -value.
5. Expand the node with the lowest $\|f\|$ -value.
6. Ignore obstacles and already visited nodes.
7. Repeat the process until the **Goal (G)** node is reached.
8. Trace back the parent nodes to obtain the optimal path.
9. Calculate the total path cost by adding the cost of each move.

Node Expansion and Priority Queue

Step Expanded Node g h f Priority Queue

1	S (1,1)	0 8 8 (2,1)[8], (1,2)[8]
2	(2,1)	1 7 8 (1,2)[8], (2,2)[8]
3	(1,2)	1 7 8 (2,2)[8]
4	(2,2)	2 6 8 (3,2)[8]
5	(3,2)	3 5 8 (3,3)[8]
6	(3,3)	4 4 8 (3,4)[8], (4,3)[8]
7	(3,4)	5 3 8 (4,3)[8], (2,4)[10]

Step Expanded Node g h f Priority Queue

8	(4,3)	5 3 8 (5,3)[8], (2,4)[10]
9	(5,3)	6 2 8 (5,4)[8], (2,4)[10]
10	(5,4)	7 1 8 G(5,5)[8]
11	G (5,5)	8 0 8 —

Optimal Path

$\boxed{S \rightarrow (2,1) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (3,3) \rightarrow (4,3) \rightarrow (5,3) \rightarrow (5,4) \rightarrow G}$

Result

The robot reaches the goal using the shortest possible path of **8 moves**. Since each move has a cost of 1, the **optimal path cost is 8**.

[Optimal Cost=8]

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

ANSWER:

a) Analysis of Constraints

Constraint	Effect on Decision-Making
Movement: Up, Down, Left, Right	Robot has four possible actions from each cell.
Obstacles	Robot must detect blocked cells and avoid them.
Limited sensing	Robot can only observe adjacent cells, so it cannot know the complete environment in advance.
Normal movement cost = 1	Robot prefers shorter paths when other conditions are equal.
Risky-zone cost = 2 extra	Entering a risky cell costs 3 in total (1 movement + 2 risk cost), so the robot may choose a longer but safer path.
Dynamic knowledge	Robot must update its map after discovering new obstacles or risky zones.
Goal = Survivor (G)	The robot must reach the survivor while minimizing total cost and maintaining safety.

b) Solution Approach – Uniform Cost Search with Online Search

Algorithm

1. Start at position **S** with cost $g(S)=0$.
2. Observe the adjacent cells using the robot's sensors.
3. Add all safe and unexplored neighboring cells to the priority queue.
4. Calculate the cost of each movement:
 - Normal cell = **1**
 - Risky cell = **1 + 2 = 3**
5. Select the node with the **lowest accumulated cost** from the priority queue.
6. Move to the selected cell.
7. Update the robot's knowledge about obstacles and risky zones.
8. If new information changes the available path, update the priority queue.
9. Continue searching and moving until the survivor **G** is reached.
10. The path with the minimum total cost is selected.

Cost Function

$$Cost(move) = 1 + Risk \quad Cost$$

Therefore:

- Safe cell → **1**
- Risky cell → **3**
- Obstacle → **Cannot enter**

Example Priority Queue

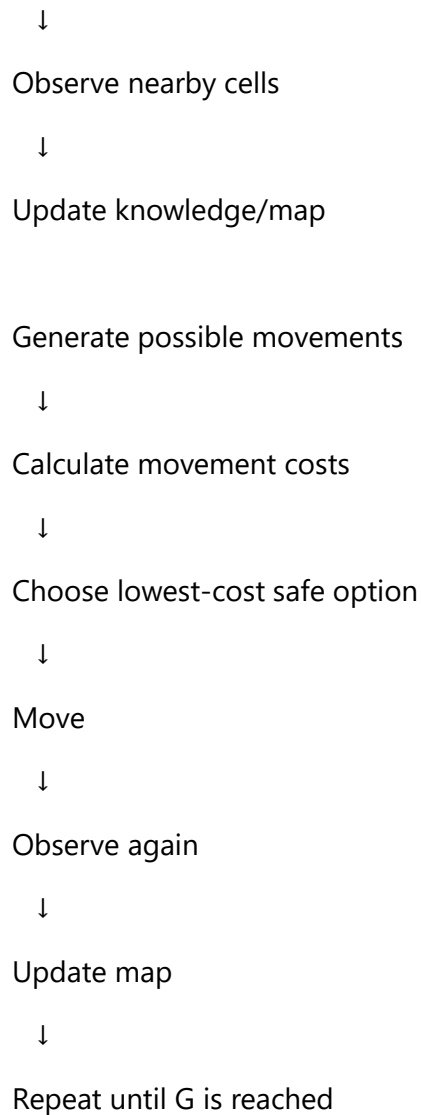
Node Path Cost Status

A	1	Safe
B	1	Safe
C	3	Risky
D	4	Safe
G	6	Goal

The robot always expands the node having the **lowest accumulated cost**.

Online Search

Start S



c) Application of AI Concepts

1. Intelligent Agent

The rescue robot acts as an **intelligent agent** because it:

- Perceives its environment using sensors.
- Processes the information received.
- Chooses suitable actions.
- Moves toward the survivor.
- Avoids obstacles and risky areas.
- Updates its knowledge as new information becomes available.

Agent: Rescue Robot

Sensors: Cameras, proximity sensors, environmental sensors

Actions: Move Up, Down, Left, Right, Stop/Rescue

Goal: Reach and rescue the survivor safely.

2. Rationality

The robot is a **rational agent** because it selects actions that maximize its performance based on the information currently available.

Its decisions consider:

- Minimum movement cost
- Risk of entering dangerous areas
- Obstacles
- Newly discovered information
- Safety of the robot
- Reaching the survivor

For example, if a short path contains a risky zone costing 3 while a slightly longer safe path costs 4, the robot can choose the safe path if safety is prioritized.

3. Environment Type

Property	Classification
Observability	Partially Observable
Determinism	Uncertain/Stochastic
Environment	Dynamic
Actions	Discrete
Number of agents	Single Agent
Knowledge	Initially Unknown/Partially Known
Task	Sequential
State Space	Grid-based

d) Justification of the Approach

Uniform Cost Search combined with an Online Search Agent is suitable because the robot has different movement costs and does not initially know the complete environment.

- UCS finds the **minimum-cost path**, not simply the shortest path.
- It handles different costs for safe and risky cells.
- The priority queue allows the robot to expand the **lowest-cost path first**.
- Online search allows the robot to **learn about obstacles and risks while moving**.
- The robot can update its decisions whenever new information is discovered.
- It is suitable for a **partially observable and uncertain disaster environment**.

Final Conclusion

The best solution is an **Online Uniform Cost Search-based intelligent agent**. The robot continuously senses its surroundings, updates its knowledge, evaluates movement costs, avoids obstacles and risky areas when appropriate, and selects the safest minimum-cost path to reach the survivor.